

Theorem: If a function $y=f(x)$ is strictly monotonic, different x are mapped onto different y

Reader: Thus, a sufficient condition for the existence of the inverse function is the strict monotonicity of the initial function. Is this right?

Author: Yes, it is.

Reader: But isn't the strict monotonicity of the initial function also a necessary condition for the existence of the inverse function?

Author: No, it is not. A one-to-one correspondence may also take place in the case of a nonmonotonic function. For example, $y = \begin{cases} 1-x, & 0 \leq x < 1 \\ x, & 1 \leq x \leq 2 \end{cases}$ (Brilliant)

Have a look at the graph of this function shown in Fig 19.

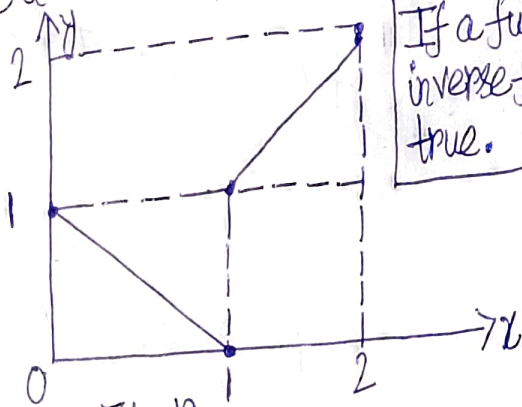


Fig. 19

If a function is strictly monotonic, it has the inverse function. However, the converse is not true. $\hookrightarrow$ V. Imp

Reader: As I understand it, in order to obtain an inverse function (when it exists), one should simply reverse the roles of x and y in the equation $y=f(x)$ defining the initial function. The inverse function will then be given by the equation $x=F(y)$. As a result, the range of the initial function becomes the domain of the inverse function.

Author: That is correct. In practice a conversion of the initial function to the inverse function can be easily performed on a graph. The graph of the inverse function is always symmetric to the graph of the initial function about a straight line $y=x$. $\hookrightarrow$ Prove it